

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ-vs-frozen CI excludes 0
9 metrics × 15 encoders · WINNER col + BLUE BOX = surgery-vs-pretrain champion duel (tie if within ε) · 10 K bootstrap

	g0-frozen	g0-pre-2X	g0-pre-enc	g0-pre-hd	g0-s3DI-enc	g0-s3DI-hd	g0-sNoDI-enc	g0-sNoDI-hd	g-pre-2X	g-pre-enc	g-pre-hd	g-s3DI-enc	g-s3DI-hd	g-sNoDI-enc	g-sNoDI-hd	WINNER
action_top1 ↑ better	38.356 ±2.264	49.973 ±2.301*	50.027 ±2.329*	38.959 ±2.219	49.260 ±2.274*	50.247 ±2.301*	49.644 ±2.329*	48.712 ±2.301*	48.219 ±2.274*	46.411 ±2.274*	42.740 ±2.329*	47.781 ±2.274*	46.466 ±2.283*	46.630 ±2.301*	47.123 ±2.247*	ViT-g·2.0 surgery 3stage_DI head
motion_cos ↑ better	0.008 ±0.001	0.228 ±0.006*	0.113 ±0.004*	0.009 ±0.001*	0.121 ±0.004*	0.119 ±0.004*	0.124 ±0.004*	0.126 ±0.004*	0.044 ±0.002*	0.042 ±0.002*	0.011 ±0.001*	0.044 ±0.002*	0.050 ±0.002*	0.057 ±0.002*	0.063 ±0.002*	ViT-g·2.0 pretrain_2X encoder
taxonomy_f1 ↑ better	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	tie
future_mse ↓ better	1.645 ±0.002	1.784 ±0.003*	1.756 ±0.002*	1.646 ±0.002	1.718 ±0.002*	N/A	1.715 ±0.002*	N/A	0.639 ±0.001*	0.632 ±0.001*	0.637 ±0.001*	0.628 ±0.001*	0.632 ±0.001*	0.636 ±0.001*	0.632 ±0.001*	ViT-g surgery 3stage_DI encoder
rollout ↓ better	-0.008 ±0.000	-0.007 ±0.000*	-0.011 ±0.000*	-0.008 ±0.000	-0.006 ±0.000*	N/A	-0.007 ±0.000*	N/A	0.005 ±0.000*	0.005 ±0.000*	0.007 ±0.000*	0.006 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	ViT-g·2.0 pretrain encoder
causal ↓ better	1.656 ±0.002	1.776 ±0.003*	1.744 ±0.002*	1.656 ±0.002	1.721 ±0.002*	N/A	1.719 ±0.002*	N/A	0.657 ±0.002*	0.652 ±0.002*	0.655 ±0.002*	0.650 ±0.002*	0.652 ±0.002*	0.658 ±0.002*	0.652 ±0.002*	ViT-g surgery 3stage_DI encoder
tdist ↓ better	0.004 ±0.001	0.005 ±0.001*	0.001 ±0.001*	0.004 ±0.001	0.003 ±0.001*	N/A	0.004 ±0.001	N/A	0.009 ±0.000*	0.008 ±0.000*	0.007 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	ViT-g·2.0 pretrain encoder
teacher_free ↓ better	0.023 ±0.000	0.032 ±0.000*	0.025 ±0.000*	0.023 ±0.000	0.042 ±0.001*	N/A	0.039 ±0.001*	N/A	0.039 ±0.001*	0.038 ±0.001*	0.044 ±0.001*	0.040 ±0.001*	0.038 ±0.001*	0.037 ±0.001*	0.038 ±0.001*	ViT-g·2.0 pretrain head
maskratio ↓ better	0.052 ±0.001	0.070 ±0.001*	0.062 ±0.001*	0.052 ±0.001	0.060 ±0.001*	N/A	0.061 ±0.001*	N/A	0.064 ±0.001*	0.065 ±0.001*	0.065 ±0.001*	0.066 ±0.001*	0.065 ±0.001*	0.062 ±0.001*	0.065 ±0.001*	ViT-g·2.0 pretrain head